

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z2,Z4

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple **Z2 Z4:** $\frac{Z_2Z_4g_m+Z_4}{2Z_2g_m+2}$

$$H(z) = \frac{Z_2Z_4g_m + Z_4}{2Z_2g_m + 2}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, R_2, \infty, R_4, \infty, \infty)$

$$H(s) = \frac{R_4}{2}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, R_2, \infty, \frac{1}{C_4s}, \infty, \infty\right)$

$$H(s) = \frac{1}{2C_4s}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, R_2, \infty, \frac{R_4}{C_4R_4s+1}, \infty, \infty\right)$

$$H(s) = \frac{R_4}{2C_4R_4s + 2}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty, R_2, \infty, R_4 + \frac{1}{C_4s}, \infty, \infty\right)$

$$H(s) = \frac{R_2g_m + s(C_4R_2R_4g_m + C_4R_4) + 1}{s(2C_4R_2g_m + 2C_4)}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \infty, R_4, \infty, \infty \right)$$

$$H(s) = \frac{R_4}{2}$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{1}{2C_4 s}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{R_4}{2C_4 R_4 s + 2}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_4 R_4 s^2 + g_m + s(C_2 + C_4 R_4 g_m)}{2C_2 C_4 s^2 + 2C_4 g_m s}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \infty, R_4, \infty, \infty \right)$$

$$H(s) = \frac{R_4}{2}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{1}{2C_4 s}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \infty, \frac{R_4}{C_4 R_4 s + 1}, \infty, \infty \right)$$

$$H(s) = \frac{R_4}{2C_4 R_4 s + 2}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \infty, R_4 + \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{C_2 C_4 R_2 R_4 s^2 + R_2 g_m + s(C_2 R_2 + C_4 R_2 R_4 g_m + C_4 R_4) + 1}{2C_2 C_4 R_2 s^2 + s(2C_4 R_2 g_m + 2C_4)}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \infty, R_4, \infty, \infty \right)$$

$$H(s) = \frac{R_4}{2}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \infty, \frac{1}{C_4 s}, \infty, \infty \right)$$

$$H(s) = \frac{1}{2C_4 s}$$

9.15 **X-INVALID-ORDER-15** $Z(s) = \left(\infty, \ R_2 + \frac{1}{C_2 s}, \ \infty, \ \frac{R_4}{C_4 R_4 s + 1}, \ \infty, \ \infty \right)$

$$H(s) = \frac{R_4}{2C_4R_4s + 2}$$

9.16 **X-INVALID-ORDER-16** $Z(s) = \left(\infty, \ R_2 + \frac{1}{C_2 s}, \ \infty, \ R_4 + \frac{1}{C_4 s}, \ \infty, \ \infty \right)$

$$H(s) = \frac{g_m + s^2 \left(C_2 C_4 R_2 R_4 g_m + C_2 C_4 R_4 \right) + s \left(C_2 R_2 g_m + C_2 + C_4 R_4 g_m \right)}{2C_4g_ms + s^2 \left(2C_2C_4R_2g_m + 2C_2C_4 \right)}$$

10 **X-INVALID-WZ**

11 **X-PolynomialError**